Gopika Jaideep

AI / ML Engineer | LLM Applications, RAG, Agentic Systems

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Summary

AI engineer with a Master of Artificial Intelligence who builds and ships AI applications end to end: LLM agents and RAG systems, Python and FastAPI backends on serverless AWS, and React and Vue front ends, delivered in agile teams and to production. Seeking a full-time AI/ML engineering role in Australia.

Career history

AI Specialist at Timeless Tributes AI, Melbourne

Apr 2026 to Sept 2026

Built and shipped AI-powered web tools for funeral directors, and led operations in a company-wide sprint making AI photo and video restoration the core product differentiator.

Achievements

- Built PhotoAlchemy, a production photo-restoration and animation tool (React, Vite, Replicate API, GFPGAN, Real-ESRGAN, Nano Banana)
- Built LiveFlow, a funeral-director-facing PWA for livestreamed ceremony playback
- Built an AI video-generation prompt and credit-efficiency system spanning four platforms (Runway, OpenArt, Kling and ElevenLabs)

Key responsibilities

- Coordinated architecture with an external development agency through a TypeScript monorepo contract layer
- Ran QA testing, wrote SOPs and production trackers, and drafted outreach to around 150 independent funeral directors
- Researched AI-authenticity and backlash concerns specific to the memorial industry to inform product positioning

Data Project Lead (Volunteer) at AusWise, Melbourne

Mar 2026 to Present

Promoted from Data Project Coordinator to Data Project Lead.

Key responsibilities

- Built a social media analytics pipeline (Instagram scraping via Apify, XGBoost and SHAP, BERTopic, Groq LLM summarisation) producing branded performance reports
- Manage a 7-member data team, working closely across HR, employability, public health, marketing and project management
- Manage AusWise's award-submission documentation and planned VIP invitations for the Safe and Sound fundraising event at Library at The Dock

Achievements

- Organised the AusWise Walkathon 2026 (370+ registrants), sorting participants into five mascot-based career teams and coordinating mentors and volunteers

Retail Volunteer at Australian Red Cross, Collingwood
Mar 2025 to Present

Retail volunteering: shop operations and customer service.

Data Science Intern at Verzeo
Nov 2020 to Dec 2020

Data cleaning and visualisation in Python.

Cybersecurity Intern at Verzeo
Sept 2020 to Oct 2020

Hands-on work with Kali Linux, Wireshark and Nmap.

Key projects

CareWise | Live app | Repository
Self-initiated project

AI companion for cancer caregivers, tracking symptoms, medications and appointments alongside emotional support.

- Built a six-agent orchestration layer (FastAPI, LLM-based intent routing, agent handoffs with cycle detection), implemented directly against provider SDKs (Gemini, Anthropic) without an agent framework
- Built a hybrid RAG pipeline (BM25 plus embeddings, reciprocal rank fusion) for a cited resource guide, restricting displayed citations to sources actually retrieved
- Designed layered crisis detection: deterministic keyword rules plus a parallel AI risk screen that can only add protection, with distinct handling when a carer reports risk to someone else
- Set up an eval-gated CI pipeline that blocks merges on crisis-detection or retrieval regressions, backed by 39+ Pytest tests
- Found and fixed two production bugs by testing the system in degraded states (AI layer down, compound multi-intent messages)

TraceAI | Repository | Demo
Self-initiated project

Missing-persons lead-scoring and evidence-fusion system for law enforcement and NGO caseworkers.

- Built a four-signal fusion engine (location, recency, credibility, identity) that renormalises weights for missing signals and applies a physical-plausibility cutoff
- Wrote a mutation-testing harness that deliberately breaks four safety controls (access gating, attestation, honeypot, audit chain) and checks the 48-test suite catches each
- Modelled misuse scenarios into access control (need-to-know gating, indistinguishable auth-failure responses) to close a stalking vector

Quitory | Frontend repository
Monash Master of AI industry studio project, team of 3, selected for the Monash Industry Showcase

Vaping-cessation support app on a serverless AWS backend.

- Owned six backend modules on serverless AWS (Lambda, API Gateway, RDS MySQL)

- Built a Groq-backed conversational assistant with keyword-gated topic scoping, and a hybrid rule-based and LLM health-risk calculator using the WHO Global Health Observatory API
- In the evaluation phase, selected the best of 25 prompt variants on the 75% training portion of 2,500 examples, then improved embedding-similarity relevance by 38 percentage points over a GPT-4 baseline on the held-out 25% test split
- Red-teamed with 500+ adversarial queries, with 0% unsafe outputs on the tested queries
- Built behavioural analytics for trigger frequency and peak-craving-time detection, and took over session and craving-trigger tracking from a teammate after iteration 1 to rebuild its cross-table user-data integration
- Delivered across three iterative sprints with iteration-based planning and handovers between teammates on shared modules, and contributed the About and Profile views on the Vue and Capacitor mobile frontend

Earlier projects

- **ParkingSpy (capstone):** data pipelines over 6 live data sources feeding a Random Forest model reaching 94.2% accuracy, cutting detection latency from about 8 minutes to under 30 seconds, delivered in an agile, sprint-based team
- **Network intrusion detection (B.Tech capstone):** compared Random Forest, SVM and LSTM on NSL-KDD (Random Forest 99.9% accuracy on a random 70/30 split) and built an LSTM web demo tested with simulated MQTT attacks
- **ALOKA (B.Tech project):** OpenCV-based Android accessibility app built in Java (repository)

Education

Master of Artificial Intelligence from Monash University, Melbourne

Feb 2024 to Dec 2025

- Progressed through the course's three stages: foundations (programming, computer architecture, statistics), core master's studies (machine learning, deep learning, planning and automated reasoning, multi-agent systems and collective behaviour) and an advanced-practice industry experience studio project, completed as Quitory
- One of the few Go8 universities offering a dedicated AI master's, with a scientist-practitioner emphasis on deploying production-ready models rather than theory alone

Bachelor of Technology, Computer Science and Engineering from SCMS School of Engineering and Technology (SSET), Kochi, India, affiliated to APJ Abdul Kalam Technological University (KTU)

2019 to 2023

- Built core CS foundations under the KTU curriculum: data structures, computer architecture and organisation, operating systems, database management systems, algorithm design and analysis, computer networks and software engineering, alongside an early introduction to AI and machine learning
- Completed final-year projects in network intrusion detection (NIDS) and OpenCV-based accessibility tooling (ALOKA)
- Built the C and C++ systems fundamentals still drawn on for lower-level and performance-sensitive work

Certifications	Professional Certification in Data Engineering, GIofAI Machine Learning with Python, Anaconda Building AI-Powered Chatbots, IBM Advanced Python, LinkedIn Learning
Key skills	**AI and ML:** Retrieval-augmented generation (hybrid BM25 and embeddings, reciprocal rank fusion), agentic AI (multi-agent orchestration, intent routing, agent handoffs), LLM evaluation methodology, eval-gated CI, mutation testing for safety-critical systems, PyTorch, distributed and multi-node training fundamentals, scikit-learn, XGBoost, SHAP, BERTopic, OpenCV **Languages and frameworks:** Python, FastAPI, TypeScript, React, Vue, Kotlin, Java (Android), C and C++ (systems fundamentals) **Cloud and data:** AWS (Lambda, API Gateway, RDS, EC2, S3), GCP (hands-on deployment, Gemini API), Databricks, Apache Spark and PostgreSQL (Data Engineering certification), SQL, SQLAlchemy, MySQL **APIs and tools:** Anthropic API, Gemini API, Groq, Replicate API, WHO Global Health Observatory API, Apify (web scraping), gTTS, concurrent and parallel systems (multiprocessing, async I/O, threading), Git, Pytest, CI/CD pipelines, Kali Linux, Wireshark, Nmap **Professional:** Cross-functional team leadership (7-member data team at AusWise), stakeholder coordination, safety-first product judgment, technical documentation and SOP writing, agile delivery **Spoken languages:** English, Hindi, Malayalam (fluent), Tamil (functional)
References	Available upon request